

Hall Effect

Jadyn Zeng, Rooney Cheung

Abstract

The Hall Effect describes the transverse voltage developed across a current-carrying conductor subjected to a perpendicular magnetic field. In this experiment, the Hall coefficients of silver (Ag) and chromium (Cr) were calculated by measuring Hall voltages and current at generated uniform magnetic fields. Linear fits of the electric field versus current density at fixed magnetic fields yielded Hall coefficients of $(-52.3 \pm 0.4) \times 10^{-11} \text{ m}^3/\text{C}$ for Ag and $(23.7 \pm 0.5) \times 10^{-11} \text{ m}^3/\text{C}$ for Cr. These signs indicate n-type conduction in silver and p-type conduction in chromium. Using the measured resistivities and Hall coefficients, the charge carrier densities, conductive mobilities, and drift velocities were determined. Silver exhibited higher mobility and lower resistivity than chromium, consistent with its simpler electronic structure and high conductivity. Deviations from literature values primarily arose from experimental uncertainties, especially in Hall voltage measurements and environmental conditions. Overall, the results demonstrate the Hall Effect and its utility in probing microscopic conductive properties, while highlighting differences in charge transport mechanisms between Ag and Cr.

1. Introduction

In a closed circuit, charges such as electrons with a **velocity** $\vec{v}$, under the influence of a **magnetic field** $\vec{B}$ experiences the Lorentz force, given by:

$$\vec{F}_L = e(\vec{v} \times \vec{B}) [N], \quad [1]$$

where $e = 1.602(10^{-19}) [C]$, is the **charge of an electron**.

If the charges are in a wire, the Lorentz force causes the charges to build up on the sides of the wire that is perpendicular to both the velocity and the magnetic field. The potential difference across the sides is known as the **Hall Voltage**, and this is known as the **Hall Effect**, discovered by E. H. Hall in 1879 [1].

Studying the Hall Effect is crucial to mastering its associated applications such as sensors and magnetometers [2].

The derivation of the Hall Effect involves basic circuit theory, which gives:

$$R = V/I [\Omega], \quad [2]$$

where V [Volts, V] is the **voltage**, I [A] is the **current** and R [Ω] is the **resistance**.

And for a wire with **resistivity** ρ [Ωm], **area** A [m^2], and **length** L [m], the resistance R [Ω] is given by [1]:

$$R = \frac{\rho L}{A} [\Omega], \quad [3]$$

Furthermore, assuming that electrons are what causes the current, with **charge** e (*same as Eq 1*) and **number density** n [$1/\text{m}^3$], travelling at a **constant average speed** v_d [m/s], the magnitude of the **current density**, $|\vec{j}|$ [A/m^2] is given by [1]:

$$|\vec{j}| = I/A, \quad [4a]$$

$$|\vec{J}| = nev_d, \quad [4b]$$

where Eq 4a gives the current density by definition, Eq 4b relates current density based on the assumptions above. Finally, the **Hall Coefficient**, R_H , a constant for any given material is given by [1]:

$$R_H = \frac{|\vec{E}|}{|\vec{B}||\vec{J}|} = \frac{1}{ne} [m^3/C], \text{ or equivalently, the fit function is:} \quad [5a]$$

$$|\vec{E}| = R_H |\vec{B}||\vec{J}| [V/m] \quad [5b]$$

where all quantities are defined above.

In this experiment, the Hall Effect will be demonstrated and observed between a pair of Helmholtz Coils using Silver (Ag) and Chromium (Cr) probes, the Hall Current and Hall Voltage will be obtained to best fit a line for Eq 5b to determine the Hall Coefficients. Microscopic quantities including the number density n , the drift velocity v_d , and the conductive mobility μ are then determined. Finally the assumption of negatively charged electrons being the charge carriers are re-examined to explain the conductive mechanisms of Cr and Ag.

2. Materials and Methods

2.1. Materials

- HP multimeter/
- Potentiometer
- Amperemeter
- DC Power Supply
- Coils Power Supply
- Helmholtz Coils
- Gaussmeter
- Traveling Microscope
- Interferometric photographs
- Cr and Ag Hall Probes

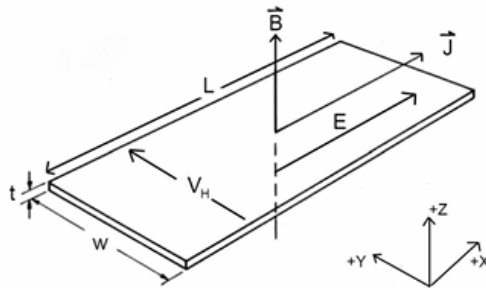
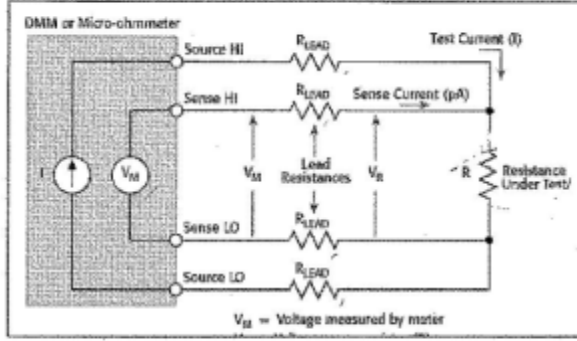


Figure 1, the hall probe geometry and its alignment with electric and magnetic fields

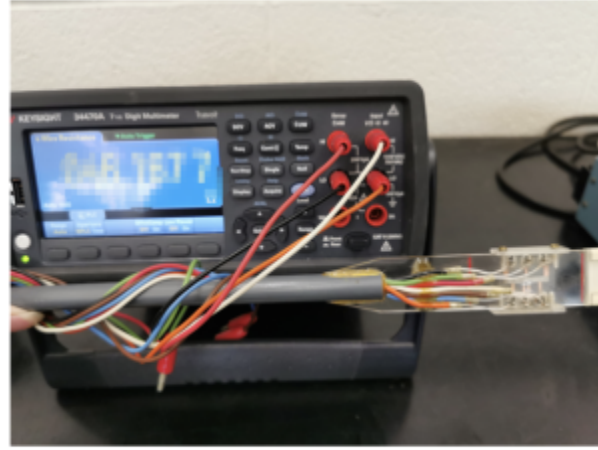
2.2. Methods

2.2.1 Measuring Resistance

Using the 4-wire (Kelvin) method, a DMM is used to measure the resistance between V_1 and V_2 [1].



(a)



(b)

Figure 2a (left), the circuit diagram of the measurement method [1]

Figure 2b (right), connection of the DMM with the probe [1]

Note that this parallel method is capable of measuring low resistances, using Eq 1, the resistance is determined by a test current passing through 1 set of wires while the voltage is measured at a second set. Here the small sense load current is neglected, i.e. $V_M \approx V_R$ for Figure 2a [1].

2.2.2 Measuring Hall Sample Dimensions

Using interferometric photographs, the following equation is applied to determine the Hall sample thickness [1],

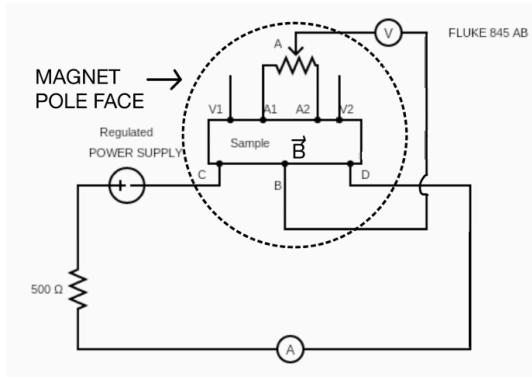
$$t = \frac{d_{step}}{d_{separation}} \times 2945 \text{ [\AA]} , \quad [6]$$

where d_{step} [mm] is the fringe step, $d_{separation}$ [mm] is the fringe separation, the 2945 Å arises from the interference fringes being observed in light of wavelength 5890 Å.

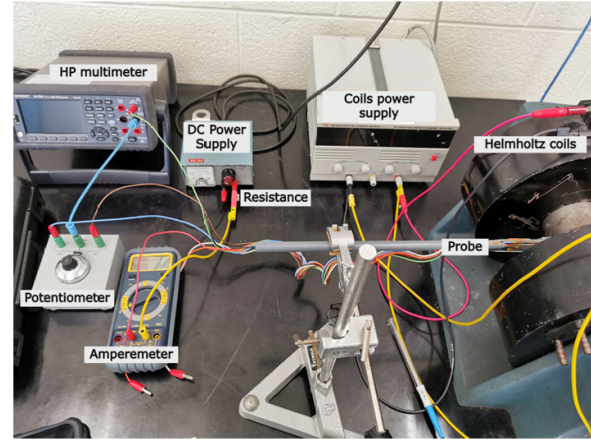
Using a travelling microscope, measure the width w and distance d between V_1 and V_2 shown in Figure 1.

2.2.3 Measuring Hall Voltage and Current

Connect the circuit as shown below in Figure 3a, 3b, make sure the limiting 500 resistance is already connected to the DC power supply. Set up the experiment as shown in Figure 3b.



(a)



(b)

Figure 3a (left), circuit diagram of Hall Probe

Figure 3b (right), setup of experiment

Turn on the DC power supply, **ensure that it never exceeds 40mA** to protect the sample. Turn on the HP multimeter and Amperemeter, check the potentiometer setting throughout the experiment. Refer to Figure 3a Use the potentiometer at A_1 and A_2 and connect the Fluke high impedance null detector between A and B. Adjust the potentiometer to the microvolt scale. Ensure the potentiometer can adjust the null even when high currents (30mA) is applied, *(it is here to remove offset potential differences between A_1 and A_2)*. Turn on the power supply for Helmholtz coils, and measure the magnetic field with a Gaussmeter by placing the flat part of the tip perpendicular to the magnetic field direction. Place the sample (*Starting with Ag*) in the center of the magnetic gap with its broad side perpendicular to the magnetic field. Measure different sets of Hall Voltage and Current, repeat this process for other magnetic fields. Repeat the above steps with the other probe (*Cr*).

3. Data and Analysis

3.1 Hall Coefficient (R_H)

Eqn 5b gives the linear fit function of the electric field strength, $|\vec{E}|$, as a function of the current density, $|\vec{j}|$, where its slope is equivalent to the product of a constant magnetic field, $|\vec{B}|$, and the Hall Coefficient. Therefore, the Hall Coefficient can be determined by the slope divided by the constant magnetic field.

The electric field strength, $|\vec{E}|$ between the parallel plates in the direction of the Hall Effect is given by:

$$|\vec{E}| = \frac{V_H}{w} [V/m] , \quad [7]$$

where V_H [V] is the Hall Voltage, w is the width of the Hall Probe [m]

$|\vec{j}|$ is given by Eqn 4a, where the y-z area (Refer to Figure 1) is given as:

$$A = t \times w [m^2] , \quad [8]$$

where w is the width of the Hall Probe [m], t is the thickness of the Hall Probe

For fitting data, refer to Appendix Table C, D for the plotted values, note that silver data is collected for magnetic fields: 392, 581, 776, 979 mT. Best fits give:

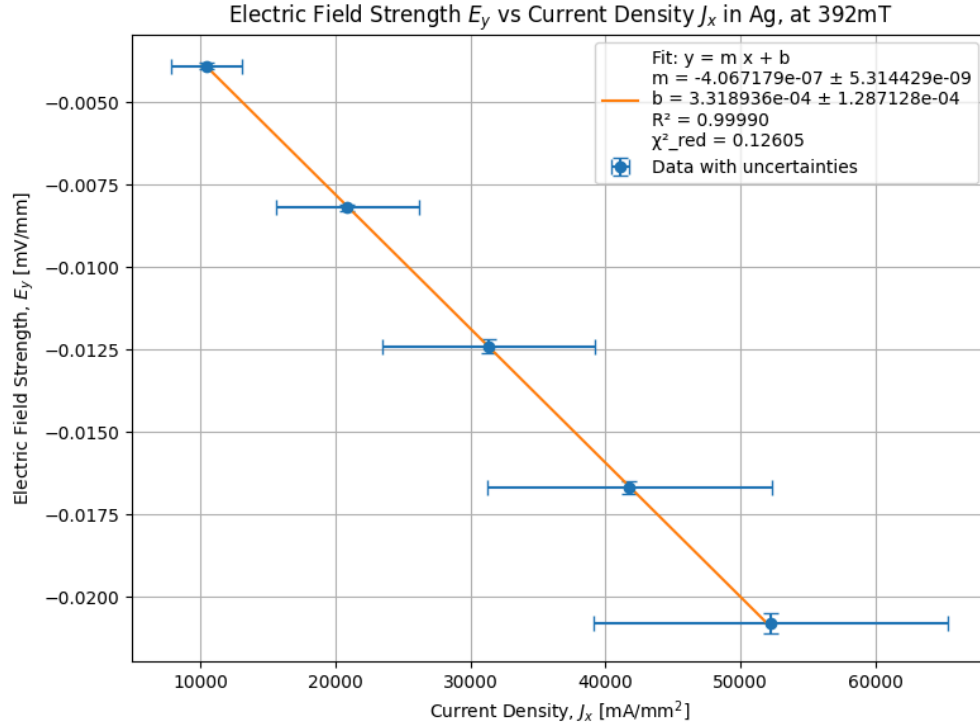


Figure 4a, best fit line of function $E_y = mJ_x + b$ at 392 mT, $m = (-4.07 \pm 0.05) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (3 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99990$, $X_{red}^2 = 0.1207$.

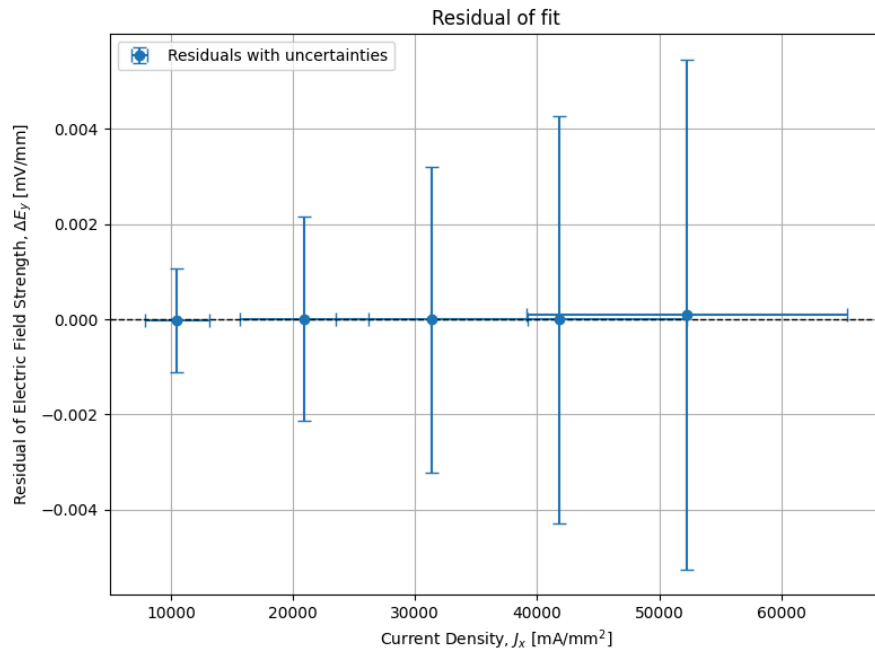


Figure 4b, Residual of the best fit line, uncertainties scale with current density due to propagation.

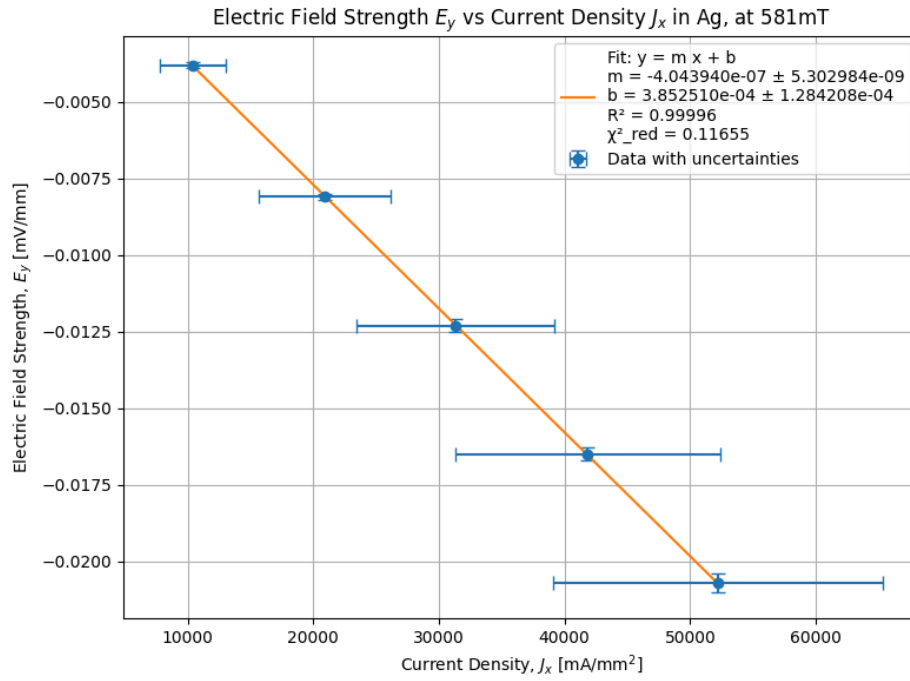


Figure 5a, best fit line of function $E_y = mJ_x + b$ at 581 mT, $m = (-4.04 \pm 0.05) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (4 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99996$, $X_{red}^2 = 0.1166$.

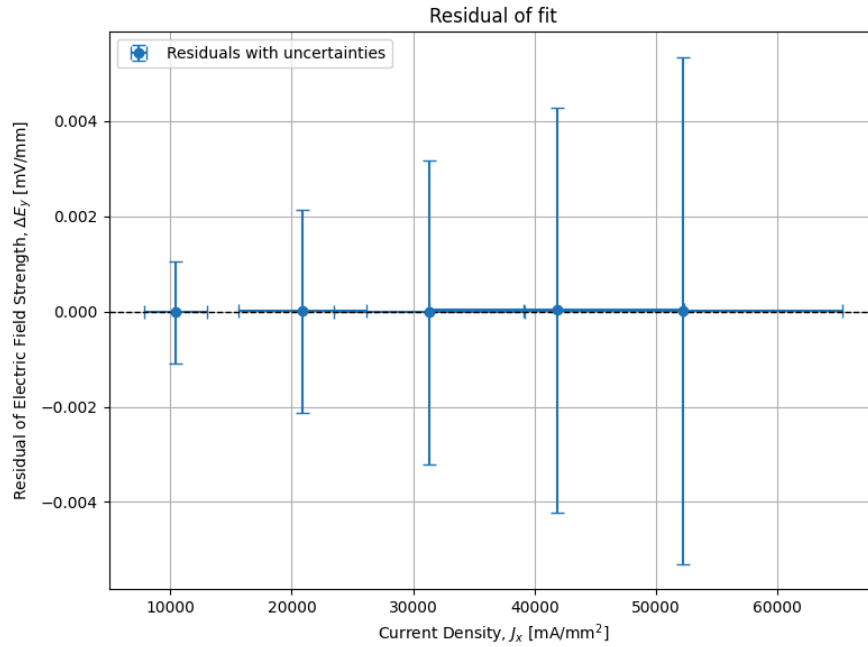


Figure 5b, Residual of the best fit line, uncertainties scale with current density due to propagation.

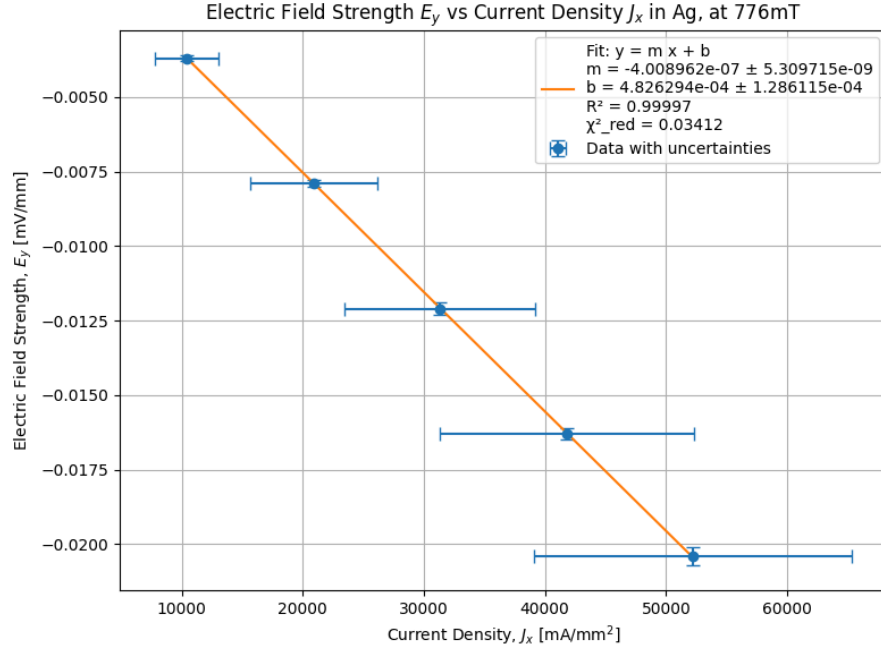


Figure 6a, best fit line of function $E_y = mJ_x + b$ at 776 mT, $m = (-4.01 \pm 0.05) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (5 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99997 \sim 1$, $X_{red}^2 = 0.03$.

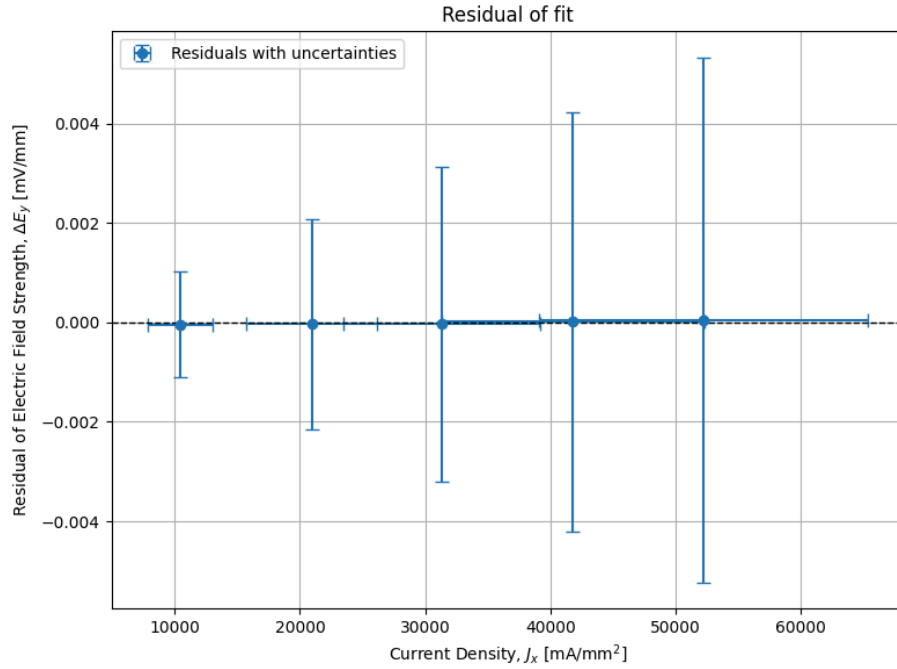


Figure 6b, Residual of the best fit line, uncertainties scale with current density due to propagation.

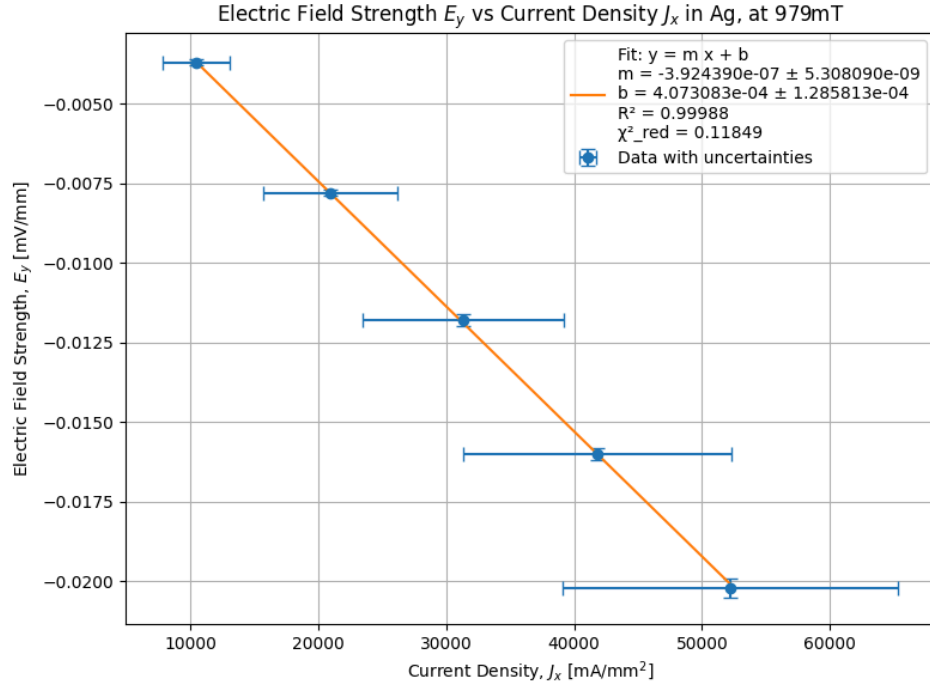


Figure 7a, best fit line of function $E_y = mJ_x + b$ at 979 mT, $m = (-3.92 \pm 0.05) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (4 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99988$, $X_{red}^2 = 0.1185$.

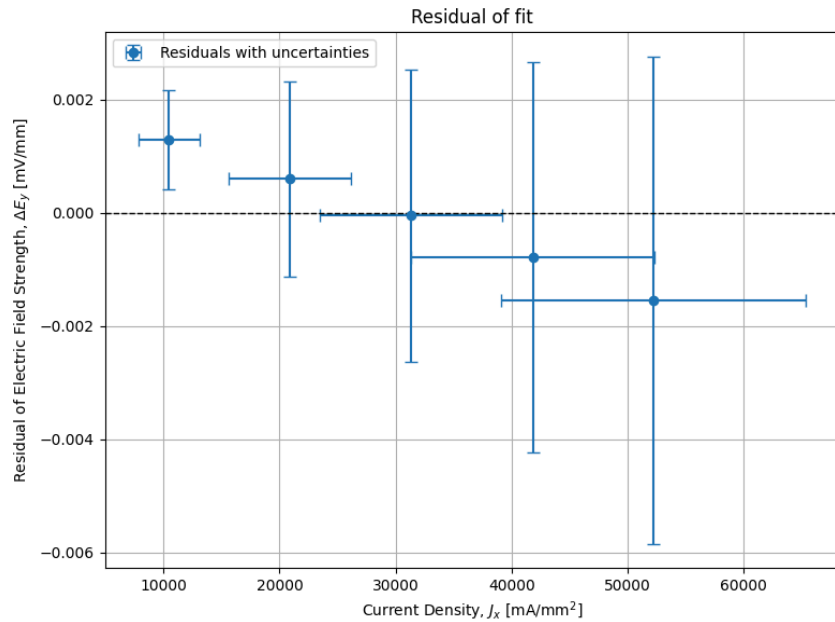


Figure 7b, Residual of the best fit line, uncertainties scale with current density due to propagation.

In general, fits show an R^2 value of approximately one, meaning that the linear line fits extremely well. In addition, the reduced chi square value, $X_{red}^2 \ll 1$, fluctuated around the range of 0.01 to 0.15. This indicates an overestimation of the uncertainties, most likely due to the fact that Electric Field Strength uncertainty scales with the Hall Voltage obtained. Furthermore, residuals are evenly distributed for 392,

581, 776 mT trials, indicating a suitable fit. However, in the 979mT trial, residuals show a linear trend, this might suggest the use of a different (quadratic for example) model for stronger magnetic fields. Upon determining each best fit slope and dividing it by the magnetic field, silver's Hall Coefficient weighted average across the tested constant magnetic fields is $(-52.3 \pm 0.4) \times 10^{-11} \text{ m}^3/\text{C}$.

Similarly, chromium data is collected for magnetic fields: 397, 596, 798, 995 mT. Best fits give:

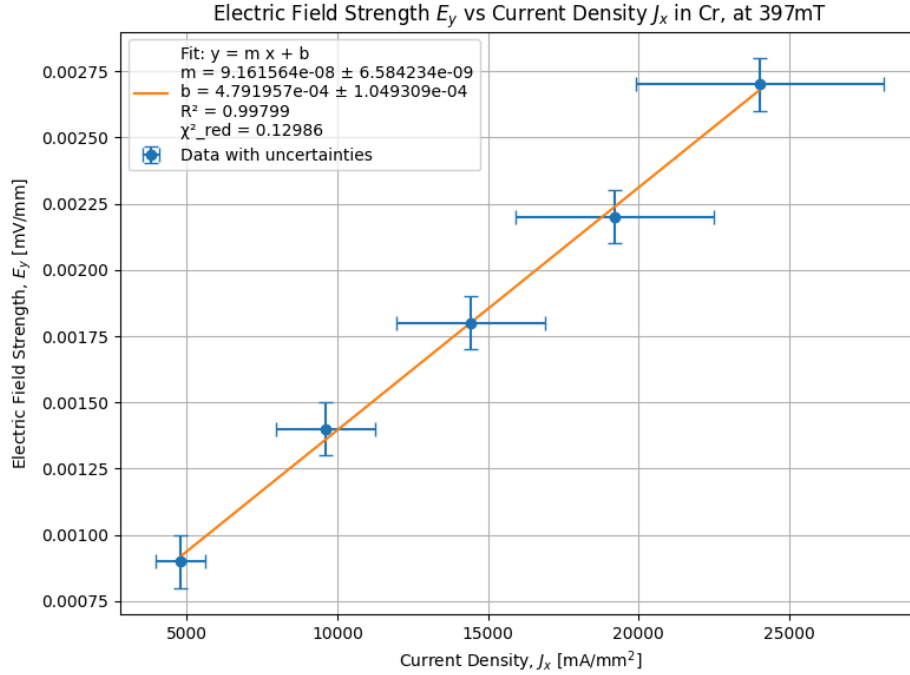


Figure 8a, best fit line of function $E_y = mJ_x + b$ at 397 mT, $m = (9.2 \pm 0.7) \times 10^{-8} \text{ m} \times \text{m}\Omega$, $b = (5 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99799$, $\chi^2_{\text{red}} = 0.1299$.

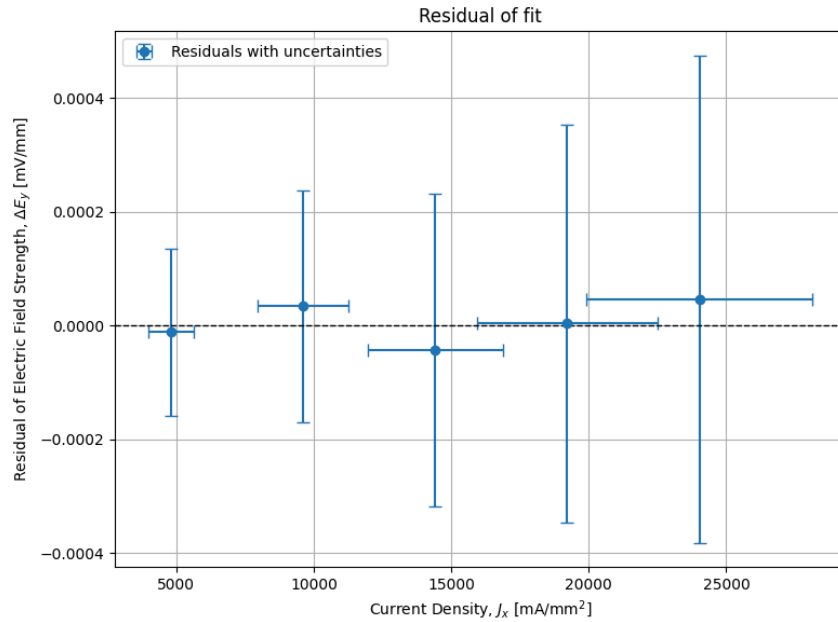


Figure 8b, Residual of the best fit line, uncertainties scale with current density due to propagation.

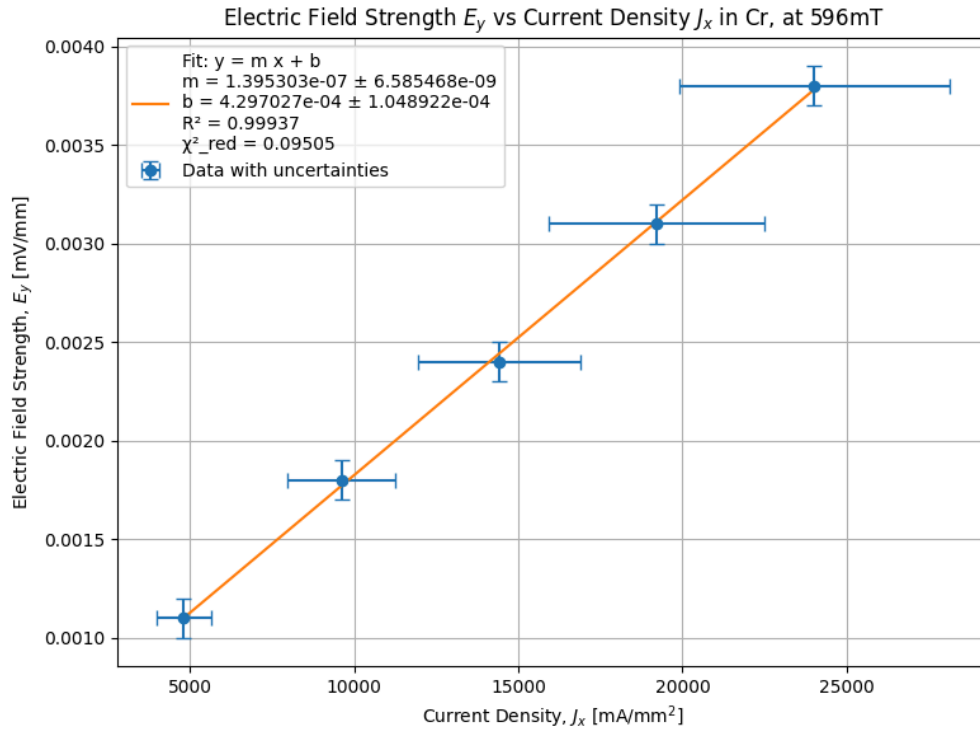


Figure 9a, best fit line of function $E_y = mJ_x + b$ at 596 mT, $m = (1.40 \pm 0.07) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (4 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99937$, $X_{red}^2 = 0.0951$.

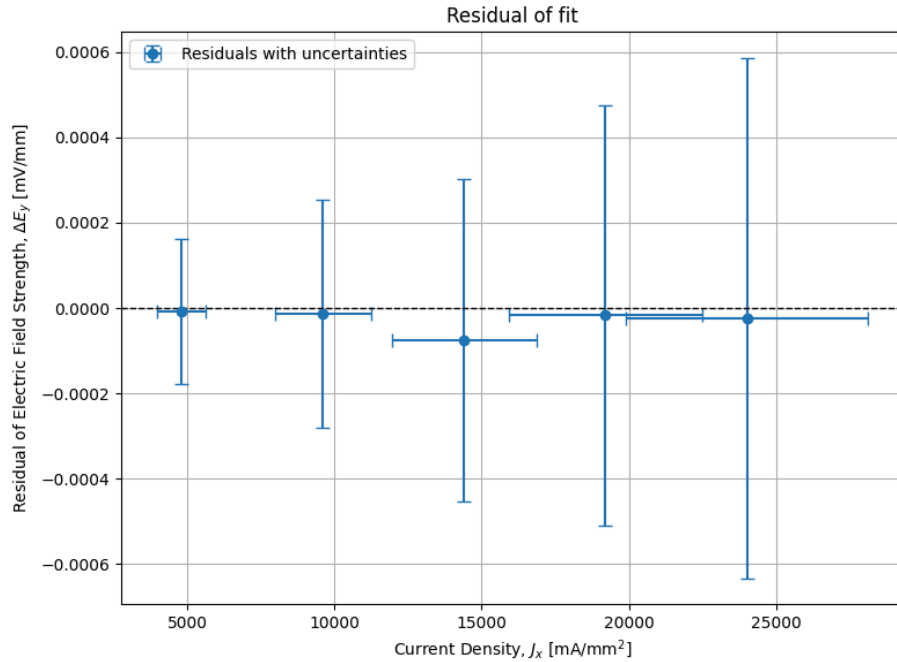


Figure 9b, Residual of the best fit line, uncertainties scale with current density due to propagation.

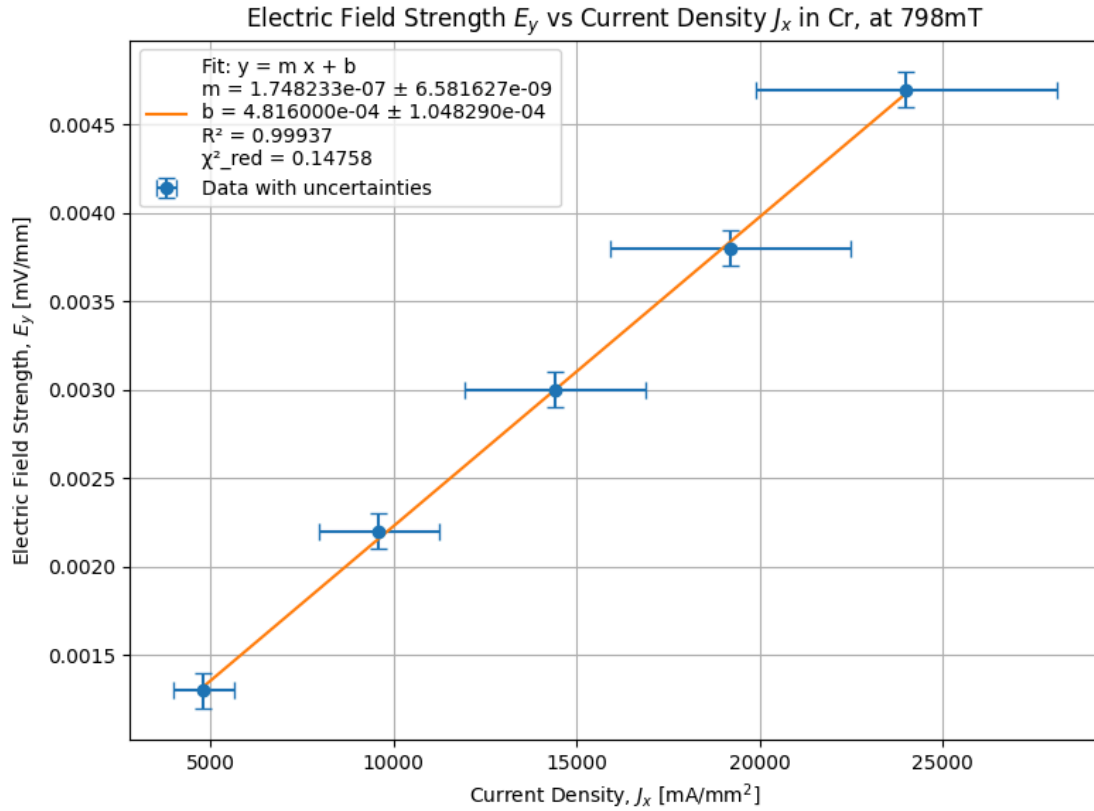


Figure 10a, best fit line of function $E_y = mJ_x + b$ at 798 mT, $m = (1.75 \pm 0.07) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (5 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99937 \sim 1$, $X_{red}^2 = 0.1476$

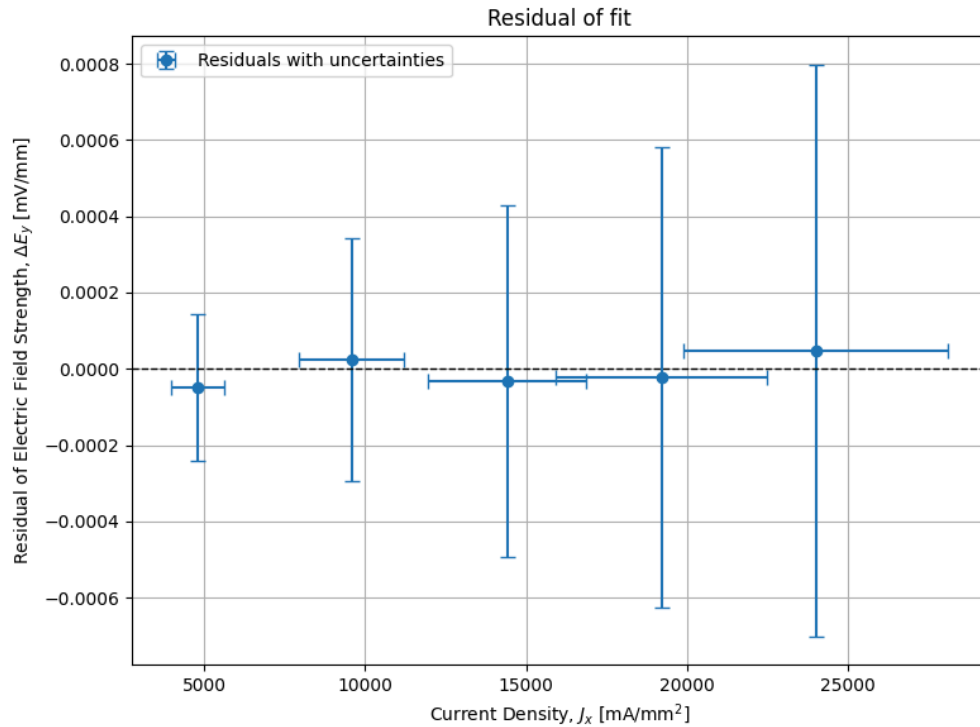


Figure 10b, Residual of the best fit line, uncertainties scale with current density due to propagation.

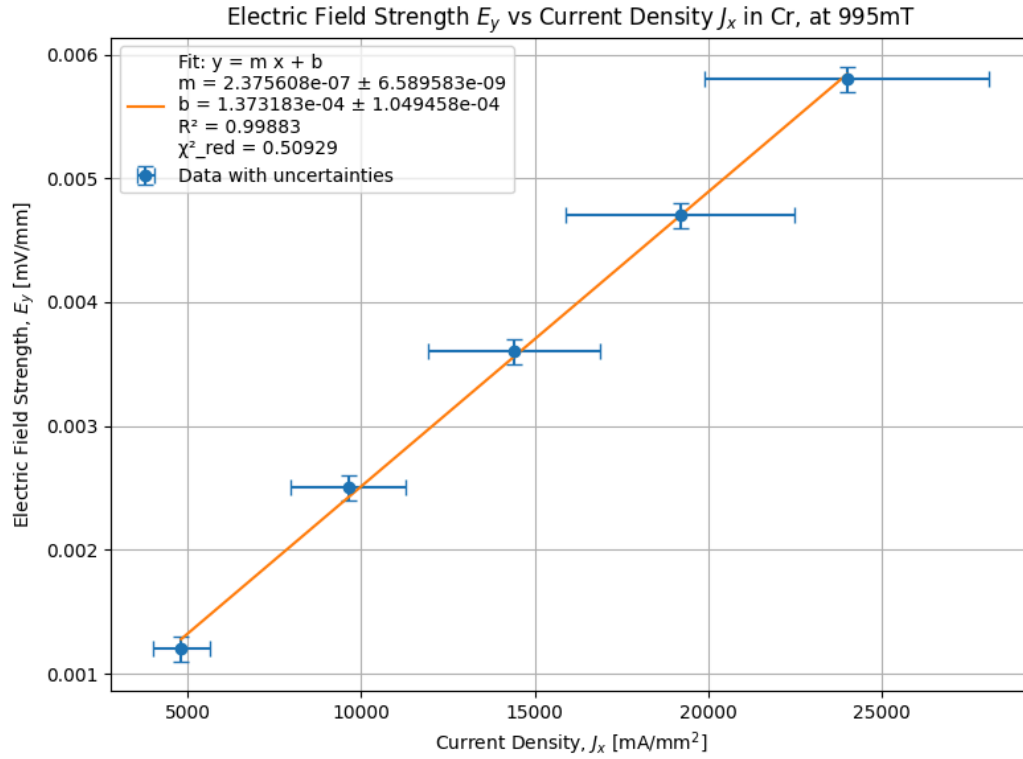


Figure 11a, best fit line of function $E_y = mJ_x + b$ at 995 mT, $m = (2.38 \pm 0.07) \times 10^{-7} \text{ m} \times \text{m}\Omega$, $b = (1 \pm 1) \times 10^{-4} \text{ mV/mm}$, $R^2 = 0.99883$, $X_{red}^2 = 0.5093$.

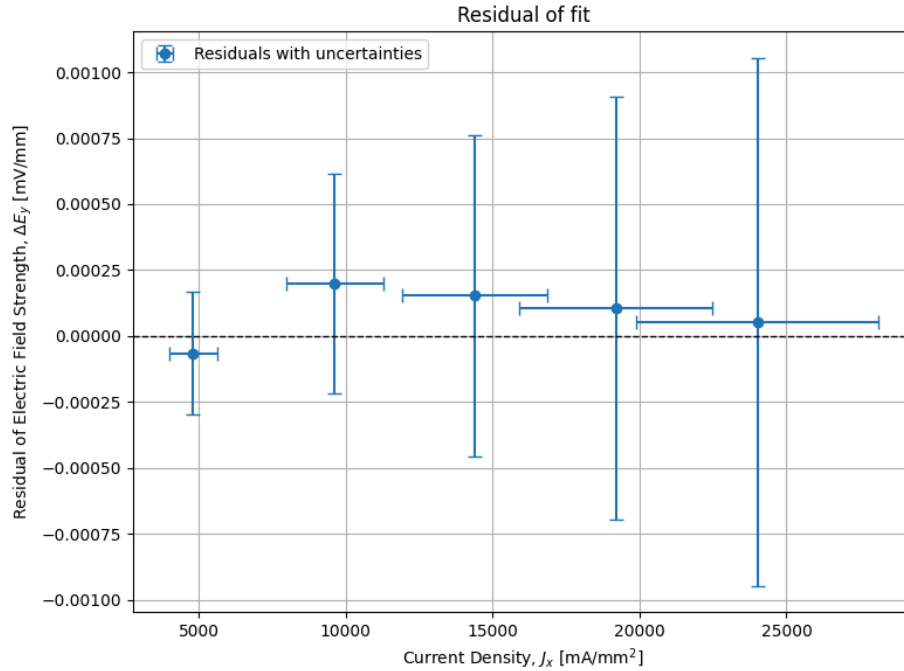


Figure 11b, Residual of the best fit line, uncertainties scale with current density due to propagation.

In general, fits show an R^2 value of approximately one, meaning that the linear line fits extremely well. In addition, for charts of 397, 596, 798 mT, the reduced chi square value, $X_{\text{red}}^2 \ll 1$, fluctuated around the range of 0.09 to 0.15. This indicates an overestimation of the uncertainties, most likely due to the fact that Electric Field Strength uncertainty scales with the Hall Voltage obtained. However, $X_{\text{red}}^2 < 1$, for the 995mT data, showing less of an overestimation. Furthermore, for 397, 798 and 995 mT, residuals are evenly distributed between positive and negative values, showing a suitable fit; for 596 mT, all residuals are <0 or ~ 0 , showing a slight underfitting. Upon determining each best fit slope and dividing it by the magnetic field, chromium's Hall Coefficient weighted average across the tested constant magnetic fields is $(23.7 \pm 0.5) \times 10^{-11} \text{ m}^3/\text{C}$.

3.2 Calculating Variables: Density of Charge Carriers, Conductive Mobility, Drift Velocity

The density of charge carriers (n), drift velocity (v_d), and conductive mobility (μ) can be expressed in terms of the Hall Coefficient (R_H), given by 3.1, and resistivity, given by Eq.3:

$$\rho = \frac{RA}{L} = \frac{R(t \times w)}{L} [\Omega m] \quad [9]$$

Refer to Appendix Table A for constant variables. This equation yields resistivity of $(3.7 \pm 0.9) \times 10^{-5} \Omega \text{ mm}$ and $(3.0 \pm 0.5) \times 10^{-3} \Omega \text{ mm}$ for silver and chromium respectively.

3.2.1 Density of Charge Carriers

$$\text{Eq 5a. gives } n = \frac{1}{R_H e} [C] \quad [10]$$

Where R_H is weighted average value calculated in *section 3.1* and e is electron's charge = $1.602 \times 10^{-19} \text{ C}$. Thus, this equation yields a charge carrier density of $(119.1 \pm 0.9) \times 10^{26} \text{ m}^{-3}$ and $(26.4 \pm 0.6) \times 10^{27} \text{ m}^{-3}$ for silver and chromium respectively.

3.2.2 Conductive Mobility

$$\text{Microscopic Ohm's Law for Conductivity } (\sigma) \text{ gives: } \sigma = ne\mu \quad [11]$$

Resistivity is defined as the reciprocal of conductivity, so

$$\rho = \frac{1}{ne\mu} [\Omega m] \quad [12]$$

$$\text{Relating resistivity to the Hall Coefficient given by Eq. 5, } \mu = \left(\frac{1}{ne}\right) \frac{1}{\rho} = \frac{R_H}{\rho} [m^2/Vs] \quad [13]$$

Thus, the conductive mobility is $(1.4 \pm 0.4) \times 10^{-5} \text{ m}^2/\text{Vs}$ and $(0.8 \pm 0.1) \times 10^{-7} \text{ m}^2/\text{Vs}$ for silver and chromium respectively.

3.2.3 Drift Velocity

Drift velocity is proportional to the electric field with constant of proportionality being conductive mobility:

$$v_d = \mu |\vec{E}| [m/s] \quad [14]$$

$$\text{From Eqn. 13, } v_d = \left(\frac{R_H}{\rho}\right) |\vec{E}| [m/s] \quad [14]$$

Drift velocity is a function of the electric field, and by extension the hall voltage.

E_y [mV/mm]	V_d [m/s]
$(3.9 \pm 0.1) \times 10^{-3}$	$(0.6 \pm 0.1) \times 10^{-4}$
$(8.2 \pm 0.1) \times 10^{-3}$	$(11.5 \pm 0.2) \times 10^{-5}$
$(12.4 \pm 0.2) \times 10^{-3}$	$(17.5 \pm 0.3) \times 10^{-5}$
$(16.7 \pm 0.2) \times 10^{-3}$	$(23.5 \pm 0.4) \times 10^{-5}$
$(20.8 \pm 0.3) \times 10^{-3}$	$(29.3 \pm 0.4) \times 10^{-5}$

Table 1: Measured electromagnetic field at 392mT for silver, and the corresponding drift velocity

E_y [mV/mm]	V_d [m/s]
$(0.9 \pm 0.1) \times 10^{-3}$	$(0.707 \pm 0.1) \times 10^{-7}$
$(1.4 \pm 0.1) \times 10^{-3}$	$(10.8 \pm 0.6) \times 10^{-8}$
$(1.8 \pm 0.1) \times 10^{-3}$	$(13.7 \pm 0.6) \times 10^{-8}$
$(2.2 \pm 0.1) \times 10^{-3}$	$(17.4 \pm 0.7) \times 10^{-8}$
$(2.7 \pm 0.1) \times 10^{-3}$	$(21.2 \pm 0.7) \times 10^{-8}$

Table 2: Measured electromagnetic field at 397mT for chromium, and the corresponding drift velocity

4. Discussion and Conclusion

Refer to *Figure 1*, the current travels in the positive x direction while the magnetic field generated by Helmholtz Coils is in the positive z direction. Thus according to *Eq 1*, the Lorentz force produced is perpendicular to both y and z, this force changes the pathway of the electron to shift up (*or down depending on the sign*), as a result, the potential difference, otherwise known as the transverse voltage between the top and the bottom of the sample is measured by the voltmeter¹.

In this experiment, the following values are obtained:

Hall Coefficient, R_H , Resistivity, ρ , Charge density, n , Drift Velocity, v_d , Conductive mobility, μ , the Hall Coefficients obtained are: $(-52.3 \pm 0.4) \times 10^{-11} \text{ m}^3/\text{C}$ for Ag; $(23.7 \pm 0.5) \times 10^{-11} \text{ m}^3/\text{C}$ for Cr, whereas literature value (Taken at room temperature) gives: -9.0×10^{-11} for Ag; $38 \times 10^{-11} \text{ m}^3/\text{C}$ for Cr [8,9]. The order of magnitude agrees with the literature value, but the deviation arises from uncertainties in the environmental conditions, such as temperature and surrounding magnetic fields. Analysis of the Hall Coefficient, R_H can be split upon the sign and the magnitude of the coefficient. The sign of the Hall

¹ Question 1

Coefficient indicates the conduction mechanism of the material. From Band Theory, conduction can be split into conduction by electrons (n-type) and conduction by “holes” (p-type): n-type conduction uses weakly bonded valence electrons as charge carriers, p-type conduction uses missing valence electrons (hence the name “holes”) as charge carriers [4]. In both cases, the charge associated, e in Eq 5a is of equivalent value, but it is **negative** for n-type conduction, **positive** for p-type conduction. Here, it indicates that silver is of n-type conduction; Chromium is of p-type conduction. Alternatively, the magnitude of the coefficient determines how dense the charge carriers are, with a greater coefficient indicating less dense charge carriers (from Eq 5a, $R_H \propto 1/n$). Thus, in this case, charge carriers are less dense in the silver sample compared to the chromium sample. The results can be summarized in the diagram below²:

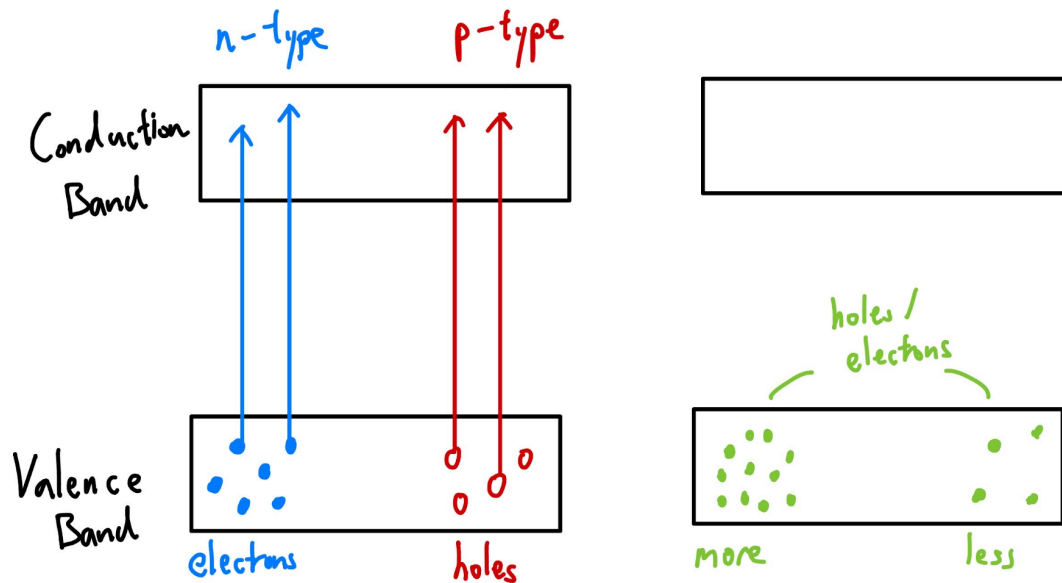


Figure 12a (left), blue indicates n-type conduction, by electrons, indicated by negative Hall Coefficient, Red indicates p-type conduction, by holes, indicated by positive Hall Coefficient.

Figure 12b (right), green indicates holes or electrons, a more dense quantity indicates smaller Hall Coefficient, a lesser quantity indicates bigger Hall Coefficient.

The resistivity measures the magnitude at which a material opposes current flow, a greater value indicates a greater opposition and vice versa. Values obtained are: $(3.7 \pm 0.9) \times 10^{-5} \Omega\text{mm}$ for Ag; $(3.0 \pm 0.5) \times 10^{-3} \Omega\text{mm}$ for Cr, whereas literature value gives: $1.47 \times 10^{-5} \Omega\text{mm}$ for Ag; $0.125 \times 10^{-3} \Omega\text{mm}$ for Cr [7, 8]. The deviation can be explained by temperature differences from standard conditions and defects/impurities in the material. Charge density obtained are: $(119.1 \pm 0.9) \times 10^{26} \text{ m}^{-3}$ for Ag; $(26.4 \pm 0.6) \times 10^{27} \text{ m}^{-3}$ for Cr. Typical values are in the range of $\sim 10^{28} \text{ m}^{-3}$, which our data falls within experimental uncertainty [3]. Drift velocity, v_d is the average velocity of free charges, it depends on the geometry of the wire, current applied and the charge density, the values obtained are in the range of 10^{-4} to 10^{-5} m/s for 10^{-7} to 10^{-8} m/s Ag; for Cr. Typically, despite the almost instantaneous propagation of

² Question 5

electrical signals, $v_d \ll c \sim 3 \times 10^8$ m/s, this is a critical result of electrical signals being transported by electromagnetic waves instead of the electrons themselves [5, 6], verified by the results above.

Conductive mobility, μ is the measurement of how easily charge carriers traverse through the material, the results obtained are: $(1.4 \pm 0.4) \times 10^{-5}$ m²/Vs for Ag; $(0.8 \pm 0.1) \times 10^{-7}$ m²/Vs for Cr. This indicates that charges move through the Ag sample more easily than they move through the Cr sample likely due to a more complex structure of Cr.

In conclusion, this experiment demonstrated the effects of Hall Effect, the transverse voltage that arises for a current over a magnetic field. A weighted, average Hall coefficient for Ag, $(-52.3 \pm 0.4) \times 10^{-11}$ m³/C and $(23.7 \pm 0.5) \times 10^{-11}$ m³/C for Cr was determined, which matches the order of magnitude for literary value. This also demonstrates Ag was of n-type conduction, Cr was of p-type conduction. In addition, values for the resistivity, charge density and drift velocity are determined for Ag and Cr respectively, demonstrating reasonable orders of magnitudes.

References

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Sample Calculations

Uncertainty Propagation

For Equation 6, $\delta|\vec{E}| = |\vec{E}|\sqrt{\left(\frac{\delta V_H}{V_H}\right)^2 + \left(\frac{\delta w}{w}\right)^2} [V/m]$

For Equation 8, $\delta A = A\sqrt{\left(\frac{\delta t}{t}\right)^2 + \left(\frac{\delta w}{w}\right)^2} [m^2]$

For Equation 9, $\delta\rho = \rho\sqrt{\left(\frac{\delta R}{R}\right)^2 + \left(\frac{\delta t}{t}\right)^2 + \left(\frac{\delta w}{w}\right)^2 + \left(\frac{\delta L}{L}\right)^2} [\Omega m]$

For Equation 10, $\delta n = \frac{1}{e} \frac{\delta R_H}{R_H^2} [C]$

For Equation 13, $\delta\mu = \mu\sqrt{\left(\frac{\delta R_H}{R_H}\right)^2 + \left(\frac{\delta\rho}{\rho}\right)^2} [m^2/Vs]$

For Equation 14, $\delta v_d = v_d\sqrt{\left(\frac{\delta\mu}{\mu}\right)^2 + \left(\frac{\delta V_H}{V_H}\right)^2 + \left(\frac{\delta w}{w}\right)^2} [m/s]$

Additional Data Tables

Table A: Constants Table

Chromium Resistance [Ω]	45.303 ± 0.0005
Chromium distance (between V1 and V2) [mm]	15.5 ± 0.25
Chromium width [mm]	16.5 ± 0.25
Chromium Fringe Step [mm]	1.5 ± 0.25
Chromium Fringe Separation [mm]	7.0 ± 0.25
Silver Resistance [Ω]	0.932 ± 0.0005
Silver distance (between V1 and V2) [mm]	12.0 ± 0.25
Silver width [mm]	19.5 ± 0.25
Silver Fringe Step [mm]	1.0 ± 0.25
Silver Fringe Separation [mm]	12.0 ± 0.25

Magnetic Field [mT]	Hall Voltage [mV]	Hall Voltage Uncertainty +/-	Hall Current [mA]	Hall Current Uncertainty +/-
392 ± 1	0.077	0.001	5.01	0.01
	0.159	0.001	10	0.01
	0.242	0.001	14.99	0.01
	0.325	0.001	20	0.01
	0.406	0.001	25.01	0.01
581 ± 1	0.075	0.001	4.99	0.01
	0.157	0.001	9.99	0.01
	0.240	0.001	15.00	0.01
	0.322	0.001	20.03	0.01
	0.404	0.001	25	0.01
776 ± 1	0.073	0.001	4.99	0.01
	0.155	0.001	10.02	0.01
	0.236	0.001	14.98	0.01
	0.317	0.001	20	0.01
	0.398	0.001	25	0.01
979 ± 1	0.072	0.001	5.00	0.01
	0.152	0.001	10.00	0.01
	0.231	0.001	15.00	0.01
	0.312	0.001	20.02	0.01
	0.393	0.001	25.00	0.01

Table B: Silver raw data table; measured hall voltage and current for different magnetic fields

Magnetic Field [mT]	Hall Voltage [mV]	Hall Voltage Uncertainty +/-	Hall Current [mA]	Hall Current Uncertainty +/-
397 ± 1	0.015	0.001	5.01	0.01
	0.023	0.001	10.01	0.01
	0.029	0.001	15.02	0.01
	0.037	0.001	20.00	0.01
	0.045	0.001	25.02	0.01
596 ± 1	0.018	0.001	5.00	0.01
	0.029	0.001	10.01	0.01
	0.039	0.001	15.00	0.01
	0.051	0.001	19.99	0.01
	0.062	0.001	25.01	0.01
798 ± 1	0.021	0.001	5.00	0.01
	0.036	0.001	9.98	0.01
	0.049	0.001	15.01	0.01
	0.063	0.001	20.01	0.01
	0.078	0.001	25	0.01
995 ± 1	0.020	0.001	5.00	0.01
	0.041	0.001	10.02	0.01
	0.059	0.001	15.00	0.01
	0.077	0.001	19.99	0.01
	0.095	0.001	25.00	0.01

Table C: Chromium raw data table; measured hall voltage and current for different magnetic fields

Magnetic Field [mT]	y-value: E_y [mV/mm]	y-uncertainty	x-value: J_x [mA/mm²]	x-uncertainty
392 $\pm$ 1	-0.0039	0.0001	10469	2630
	-0.0082	0.0001	20896	5249
	-0.0124	0.0002	31323	7868
	-0.0167	0.0002	41792	10498
	-0.0208	0.0003	52261	13128
581 $\pm$ 1	-0.0038	0.0001	10427	2619
	-0.0081	0.0001	20875	5244
	-0.0123	0.0002	31344	7873
	-0.0165	0.0002	41855	10514
	-0.0207	0.0003	52240	13122
776 $\pm$ 1	-0.0037	0.0001	10427	2619
	-0.0079	0.0001	20938	5259
	-0.0121	0.0002	31302	7863
	-0.0163	0.0002	41792	10498
	-0.0204	0.0003	52240	13122
979 $\pm$ 1	-0.0037	0.0001	10448	2625
	-0.0078	0.0001	20896	5249
	-0.0118	0.0002	31344	7873
	-0.0160	0.0002	41834	10508
	-0.0202	0.0003	52240	13122

Table D: Silver experiment plotted values to fit linear expression of Eqn 5b

Magnetic Field [mT]	y-value: E_y [mV/mm]	y-uncertainty	x-value: J_x [mA/mm²]	x-uncertainty
397 $\pm$ 1	0.0009	0.0001	4811	823
	0.0014	0.0001	9613	1645
	0.0018	0.0001	14425	2468
	0.0022	0.0001	19207	3287
	0.0027	0.0001	24028	4112
596 $\pm$ 1	0.0011	0.0001	4802	822
	0.0018	0.0001	9613	1645
	0.0024	0.0001	14406	2465
	0.0031	0.0001	19198	3285
	0.0038	0.0001	24019	4110
795 $\pm$ 1	0.0013	0.0001	4802	822
	0.0022	0.0001	9584	1640
	0.0030	0.0001	14415	2467
	0.0038	0.0001	19217	3288
	0.0047	0.0001	24009	4109
995 $\pm$ 1	0.0012	0.0001	4802	822
	0.0025	0.0001	9623	1647
	0.0036	0.0001	14406	2465
	0.0047	0.0001	19198	3285
	0.0058	0.0001	24009	4109

Table E: Chromium experiment plotted values to fit linear expression of Eqn 5b